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# Can GenAI help undergraduate students become independent writers? An intervention study on its effects on writing motivation, feedback literacy, and self-regulated learning strategies

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## ABSTRACT

This mixed-methods study adopts a longitudinal perspective to investigate whether and how GenAI-supported feedback practice influences students' writing motivation, feedback literacy, and self-regulated learning (SRL) strategy use, three aspects essential for developing as independent writers. Participants were 61 Chinese undergraduates enrolled in an academic writing program. Quantitative and qualitative findings revealed a differentiated pattern of change across the three domains: writing interest increased, while self-efficacy showed only slight gains; feedback literacy improved in appreciating feedback, managing affect, and taking action, but showed limited improvement in acknowledging feedback sources and making judgments; and SRL strategy use improved in text-generating and revising, with modest gains in planning but a slight decline in self-monitoring. Together, these findings provide nuanced insights into how GenAI-mediated feedback settings can both support and constrain students' development as independent writers, highlighting the need for pedagogical scaffolding and ethical guidance.

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Generative artificial intelligence; writing motivation; feedback literacy; self-regulated learning strategy use

## Introduction

Feedback is increasingly recognised as a critical component in writing instruction. With the growing integration of generative Artificial Intelligence (GenAI) into feedback practices, learners may become more actively engaged in writing activities (e.g. Yan & Zhang, 2024; Yeung, 2025). At the same time, concerns have emerged regarding potential overreliance on GenAI-generated feedback and its implications for learners' responsibility in managing their writing processes (e.g. Chen et al., 2024; Lu et al., 2026).

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In light of these considerations, it is essential to examine whether GenAI-supported feedback practice could genuinely support students' broader development as independent writers. Independent writers are generally understood as those who assume increasing responsibility and exercise autonomous control over their writing processes and development (e.g. Hyland & Hyland, 2006; Ratminingsih et al., 2018). This perspective shifts attention from the immediate pedagogical utility of GenAI to its potential formative role in fostering sustained learner independence in writing.

Writer independence, in this sense, concerns learners' autonomy and responsibility in directing their own writing development. These qualities are enacted through several complementary, process-oriented capacities, particularly motivation to write (e.g. Hidi & Boscolo, 2006), the ability to effectively utilise feedback (e.g. Hyland & Hyland, 2006), and the capacity to self-regulate the writing process (e.g. Zimmerman & Risemberg, 1997).

These capacities take on renewed significance when GenAI becomes embedded in writing instruction. In technology-rich environments, writer independence may be understood as learners' sustained capacity to exercise autonomy and responsibility while engaging with powerful technological assistance (e.g. Dang, 2025). When certain cognitive and evaluative functions (e.g. identifying problems and assessing text quality) are partially externalised to GenAI, sustaining writer independence can be particularly demanding for second language (L2) learners. Compared with L1 writers, L2 learners face additional linguistic and cognitive demands (e.g. limited linguistic resources for expressing nuanced meanings; Lu, Zhu, et al., 2024), which may increase their reliance on technological support.

Under these conditions, sustaining writer independence may require more deliberate and sustained effort in L2 contexts. More specifically, it may require learners to remain sufficiently motivated to take ownership of their work (e.g. Zhang et al., 2025), to critically engage with feedback rather than accept it unreflectively (e.g. Zhan & Yan, 2025), and to actively regulate writing and revision processes instead of delegating control to technology (e.g. Mohebbi, 2025). Writer independence, therefore, is not reducible to any single attribute. Rather, it reflects the coordinated development of these complementary, process-oriented capacities. It is therefore conceptually distinct from writing quality, which represents a product-oriented outcome rather than the underlying processes that underpin writer independence.

Derived from these core characteristics, writing motivation, feedback literacy, and self-regulated learning (SRL) strategy use serve as central aspects for understanding how learners, especially L2 learners, may develop as independent writers in the context of GenAI-supported feedback practice.

Although current research has begun to explore the effects of GenAI on these constructs, existing studies remain limited in scope and fragmented in focus. Most examine individual variables in isolation or treat them as undifferentiated constructs without attending to their specific dimensions (e.g. Apriani et al., 2024; Liu et al., 2024; Rad et al., 2024). Few studies have considered these constructs together within a shared conceptual framing of independent writer development in GenAI-supported feedback settings.

Guided by the lens of writer independence, the present study investigates whether and how students' writing motivation, feedback literacy, and SRL strategy use change

during a ten-week GenAI-supported feedback intervention. It further explores students' perceptions of this feedback experience through a mixed-methods approach, capturing both observable developmental changes and lived experiences. By situating these changes within an authentic instructional context, the study aims to deepen understanding of the formative role of GenAI-supported feedback practice and offer pedagogical insights for supporting independent writer development.

## Literature review

### *GenAI-supported feedback practice and writing motivation*

Writing motivation refers to learners' dispositions, beliefs, and goals shaped through their experiences with writing across tasks and contexts (Boscolo & Gelati, 2019). It is a central driver of writing development (e.g. Camacho et al., 2021; Troia et al., 2012) and is commonly conceptualised as a multidimensional construct that includes, among other components, self-efficacy and writing interest (Graham, 2018).

Self-efficacy reflects writers' judgments of their capability to perform writing tasks and their confidence in executing writing activities (e.g. Sun & Wang, 2020), whereas writing interest refers to the extent to which writers find the writing process engaging and worth investing effort in, e.g. Graham (2018). Together, these dimensions capture both competence ('can I do it?') and willingness ('do I want to do it?'), which are central to learners' development as independent writers.

Previous studies have reported positive associations between various forms of feedback (e.g. teacher, peer, and automated writing evaluation [AWE]) and L2 students' writing self-efficacy and interest (e.g. Fan & Xu, 2020; Ruegg, 2018; Sari & Han, 2024). However, research on GenAI-supported feedback practice remains limited. Preliminary evidence suggests that interacting with such tools may enhance students' writing self-efficacy (e.g. Apriani et al., 2024; Huang & Mizumoto, 2024).

At the same time, some scholars caution that the motivational effects of GenAI-supported feedback practice may be multifaceted, potentially exerting both facilitating and inhibiting influences (e.g. Rodríguez-Ruiz et al., 2025). This suggests that GenAI-supported feedback practice may affect different motivational dimensions in distinct ways, highlighting the need for more fine-grained investigation into how it shapes motivational processes in writing.

Self-Determination Theory (SDT; Ryan & Deci, 2000, 2020) provides a useful lens for understanding how motivational processes may operate in the context of GenAI-supported feedback practices. SDT posits that autonomy (a sense of volition and self-direction), competence (a sense of mastery), and relatedness (a sense of connectedness with others) are three essential psychological needs that underpin intrinsic motivation. This framework helps illuminate how the satisfaction or frustration of these needs may shape learners' motivational beliefs, including self-efficacy and interest.

From an SDT perspective, perceived competence, a core psychological need, is theoretically associated with learners' self-efficacy beliefs, as experiences of mastery and improvement are often linked to stronger beliefs in one's capacity to succeed (e.g. Li et al., 2022). GenAI-supported feedback practice may shape such

competence-related experiences in contrasting ways. On the one hand, by offering immediate and supportive information that helps students refine their writing (e.g. Barrot, 2023), it may enhance learners' perceived competence, which may correspond with stronger self-efficacy beliefs. On the other hand, excessive reliance on GenAI-generated suggestions may limit opportunities for independent mastery experiences (e.g. Rodríguez-Ruiz et al., 2025), potentially corresponding with weaker self-efficacy beliefs.

Writing interest may be understood in relation to autonomy need satisfaction within SDT. Interest reflects learners' willingness to engage in a task for its own sake and is often associated with experiences of personal choice and self-direction in the learning process (e.g. Graham, 2018; Hidi et al., 2023). In GenAI-supported feedback contexts, students may experience greater autonomy when they are able to decide whether and how to incorporate feedback into their revisions, thereby perceiving a sense of control and ownership over their writing (e.g. Zhang et al., 2025). Such autonomy-supportive experiences may be associated with heightened intrinsic interest in writing.

In addition to autonomy, satisfaction of the need for relatedness may also be relevant to understanding learners' writing interest within SDT. Prior studies suggest that GenAI tools can provide socio-emotional feedback, such as praise, encouragement, or empathetic responses, that helps students feel acknowledged and supported throughout the writing process (e.g. Lu et al., 2024). These experiences of perceived relatedness may contribute to more positive emotional engagement with writing tasks and sustained interest over time.

Taken together, from an SDT perspective, GenAI-supported feedback practice may shape learners' motivational beliefs, including self-efficacy and writing interest, through experiences related to the satisfaction or frustration of autonomy, competence, and relatedness needs. However, the ways in which these processes unfold may vary across motivational dimensions, highlighting the need for further inquiry.

### ***GenAI-supported feedback practice and feedback literacy***

Feedback literacy is widely regarded as a crucial competence that enables students to benefit from feedback practices, encompassing the knowledge, skills, and dispositions required to interpret and use feedback effectively (e.g. Carless & Boud, 2018). In writing contexts, Yu et al. (2022) conceptualised feedback literacy as students' capacities to appreciate, evaluate, and utilise writing feedback while managing their emotions during the process. Their framework includes five dimensions: appreciating feedback (valuing its importance), acknowledging different feedback sources (recognising diverse providers), making judgments (evaluating feedback quality and relevance), managing affect (regulating emotional responses), and taking action (applying feedback for improvement).

The present study adopts Yu et al.'s (2022) conceptualisation and corresponding L2 scale for three reasons. First, the scale has been empirically validated and aligns with prior work emphasising the multidimensional nature of feedback literacy (e.g. Carless & Boud, 2018). Second, it was developed and validated in Chinese university academic writing contexts, closely matching the instructional setting and participant

population of this study. Third, the five dimensions capture learners' process-oriented capacities to appreciate, evaluate, manage, and act on feedback, capacities that remain necessary even when feedback is generated by GenAI. Recent studies in GenAI-supported writing contexts (e.g. Liu et al., 2026) have also adopted this scale, supporting its applicability in technology-enhanced feedback environments.

Existing studies suggest that different forms of feedback, including teacher, peer, and self-feedback, may influence the development of students' feedback literacy in different ways (e.g. Hoo et al., 2022; Weng et al., 2024). For example, Weng et al. (2024) found that peer feedback improved students' ability to recognise and judge feedback but had limited effects on understanding feedback sources, emotion regulation, and feedback utilisation.

However, few studies have examined the potential impact of GenAI-supported feedback practice on L2 students' feedback literacy. Rad et al. (2024) provide some indirect evidence: in an 11-week intervention with 46 university students, those using the AI tool Wordtune showed greater gains in overall feedback literacy than a control group. Yet this study did not examine specific literacy dimensions, and the tool differs from GenAI systems that provide real-time, adaptive dialogue rather than fixed feedback.

From a socio-constructivist perspective, learning develops through interaction and dialogue, where learners negotiate meaning and construct understanding collaboratively (e.g. Hickey, 1997; Rannikmäe et al., 2020). In writing feedback contexts, opportunities to discuss, interpret, and respond to feedback are therefore central to developing feedback literacy (e.g. De Kleijn, 2023). With its real-time, adaptive feedback capabilities (e.g. Barrot, 2023), GenAI may provide such dialogic opportunities, potentially shaping how learners understand, evaluate, and use feedback, as well as how they regulate their emotional responses. Further research is therefore needed to examine whether and how GenAI-supported feedback practice influences different dimensions of students' feedback literacy in writing.

### ***GenAI-supported feedback practice and SRL strategy use***

Self-regulated learning (SRL) refers to the processes through which learners set goals, monitor progress, and adjust strategies to achieve desired outcomes (e.g. Panadero, 2017; Schunk & Zimmerman, 2011). In writing, SRL is reflected in strategies used across the stages of writing – prewriting, during writing, and postwriting – including planning (e.g. setting goals and organising ideas), generating text (drafting content), self-monitoring (checking progress and detecting problems), and revising (refining text) (e.g. Bai et al., 2022). These strategies reflect writers' ability to manage cognitive and metacognitive processes throughout writing.

Previous research shows that feedback is an important external factor shaping how L2 students deploy SRL strategies (e.g. Tian et al., 2022; Zhu et al., 2024). For example, Tian et al. (2022) found that students invested greater effort in revising strategies when responding to automated feedback than to peer or teacher feedback, possibly because automated feedback often focuses on surface-level features and therefore requires additional interpretation and integration. Such findings suggest that students adapt their SRL strategies in response to the characteristics of the feedback they receive.

Given that GenAI-generated feedback is real-time, adaptive, and content-rich (e.g. Lu et al., 2024; Yao et al., 2025), students may further adjust their SRL strategies when engaging with this form of feedback. Although direct evidence remains limited, Liu et al. (2024) reported improvements in primary school L2 students' use of planning, text generation, self-monitoring, and revising strategies following GenAI-mediated writing instruction compared with conventional instruction. While GenAI feedback was only one component of the intervention, these findings suggest that GenAI's feedback affordances may shape how students regulate their writing processes.

However, whether and how GenAI-supported feedback practice influences SRL strategy use in L2 writing remains underexplored, warranting more fine-grained and theoretically informed investigation. Because SRL involves learners' active management of cognitive and metacognitive processes (e.g. Panadero, 2017), GenAI may play a dual role. On the one hand, it can provide timely, adaptive cues across writing stages (e.g. Rong et al., 2025), potentially supporting key SRL sub-strategies. For instance, during planning it may help clarify task demands and goals; during text generation it may prompt idea elaboration or improve coherence; during self-monitoring it may cue learners to check goal-text alignment; and during revision it may facilitate both surface editing and higher-level restructuring (e.g. Chen et al., 2024; Li et al., 2025; Liu et al., 2024).

On the other hand, excessive reliance on GenAI guidance may reduce learners' initiative for independent regulation throughout the writing process. Learners may defer goal-setting and decision-making to GenAI suggestions, incorporate generated revisions without critically evaluating relevance, or engage less in internal monitoring, patterns consistent with what Fan et al. (2025, p. 492) termed 'metacognitive laziness', a tendency to depend on external rather than internal regulation. Such reliance may constrain the effective deployment of SRL strategies across planning, text generation, self-monitoring, and revision over time.

### ***The present study***

The review of existing research highlights several observations. Although initial investigations have addressed the impact of GenAI-supported feedback practice on writing motivation, feedback literacy, and SRL strategy use to some extent, research on this topic remains limited and fragmented. Existing studies often examine only one variable, and when they do, they may overlook its specific dimensions (e.g. Apriani et al., 2024; Rad et al., 2024). Consequently, there is a lack of comprehensive research that simultaneously examines writing motivation, feedback literacy, and SRL strategy use from a unified perspective. One promising lens is that of fostering independent writers (e.g. Hyland & Hyland, 2006; Ratminingsih et al., 2018).

Writer independence, as noted in the Introduction, concerns learners' capacity to assume autonomy and responsibility in directing their own writing development. Such autonomy and responsibility are not merely abstract dispositions but are enacted through concrete processes in writing practice (e.g. Hidi & Boscolo, 2006; Hyland & Hyland, 2006; Zimmerman & Risemberg, 1997).

First, to write independently, learners typically need a motivational foundation to engage with writing tasks and sustain effort over time. Writing motivation, reflected in constructs such as writing self-efficacy and writing interest, captures learners' willingness and confidence to participate in and persist with writing activities.

Second, independent writing involves critical processing of external input rather than unreflective acceptance. In feedback-rich environments, especially those mediated by GenAI, learners need to interpret, evaluate, and decide when and how to use feedback. Feedback literacy reflects learners' capacity to interpret, evaluate, and use feedback constructively.

Third, autonomy and responsibility must be enacted through regulatory action during writing. Self-regulated learning (SRL) strategy use reflects the planning, monitoring, and revising processes through which learners manage their writing and coordinate feedback use as one source of information for improvement.

In this sense, writing motivation, feedback literacy, and SRL strategy capture distinct yet necessary functional aspects of independent writer development. They describe how writer independence is enacted through writing processes in practice.

To address these gaps, the present study examines whether and how students' writing motivation, feedback literacy, and SRL strategy use develop during a ten-week GenAI-supported feedback intervention in an academic English writing program. Guided by the lens of writer independence, it aims to advance understanding of independent writer development in GenAI-supported contexts and inform the design of GenAI-integrated pedagogies.

Accordingly, the study addresses the following research questions:

RQ1: Whether and how students' writing motivation, feedback literacy, and SRL strategy use change over the course of the ten-week GenAI-supported feedback practice?

RQ2: If changes are observed, how can these be interpreted through students' experiences and perceptions of the GenAI-supported feedback practice?

## Methods

### *Participants and context*

A total of 61 Chinese undergraduate students (9 males and 52 females; mean age = 20.33, SD = 0.54) participated in the study. All were enrolled in education-related majors (e.g. curriculum and instruction, comparative education, higher education) at a research university in southern China. Participants took part in a 10-week academic English writing program designed to develop genre awareness, academic writing conventions, and self-regulated writing strategies. The program consisted of weekly three-hour sessions combining instruction, guided writing activities, and feedback practice. Writing tasks were designed to consolidate skills practiced in the sessions, with GenAI-supported feedback practice serving as a core instructional component that enabled students to interact with GenAI under instructor guidance. The program was delivered by an instructor with over ten years of experience in academic writing instruction.

### *Procedures*

The study spanned the entire 10-week academic writing program. During the first two weeks, participants were introduced to the program schedule, task requirements, and evaluation criteria, and completed a pre-test survey assessing baseline levels of

writing motivation, feedback literacy, and SRL strategy use. They then received preparatory instruction on SRL strategies and training on the GenAI tool Kimi (Moonshot AI, China, October 2023; <https://kimi.moonshot.cn/>), including guidance on formulating effective prompts to obtain actionable feedback. This preparation aimed to support students' use of GenAI during subsequent feedback-and-revision activities embedded in the writing tasks (see [Appendix A](#) for detailed instructional procedures).

Throughout the ten-week program, participants completed three focal writing tasks (Weeks 3, 6, and 9). Each task involved an in-class drafting stage followed by a GenAI-supported feedback-and-revision session using Kimi. Detailed procedures for the drafting, feedback, and revision stages are provided in [Appendix A](#).

In Week 10, the program concluded with a summary session reflecting on the content covered and participants' progress. The post-test questionnaire, identical to the pre-test, was administered online during class time. Additionally, semi-structured interviews were conducted with nine participants to gather insights into their perceptions of the feedback process and writing experiences. Ethical approval was obtained from the University's Institutional Ethics Review Board prior to the study.

## ***Instruments***

### ***Three writing tasks***

Throughout the program, participants completed three integrated writing (IW) tasks. This genre was selected because it requires students to extract information from source materials, evaluate evidence, formulate ideas, and construct coherent arguments. The topics were 'Online Learning vs. Traditional Classroom Learning', 'The Impact of GenAI on Undergraduate Learning', and 'Choosing a University Major: Interest vs. Employability', all designed to relate to students' experiences and current educational issues. For each task, participants received four reading materials (approximately 500 words in total) presenting diverse perspectives from academic articles, news reports, and essays. Based on these materials, they wrote a 200-word argumentative essay within one hour.

Task performance was assessed using four key criteria: contextual awareness, citation and synthesis, opinion and argument, and organisation and expression (Cheong et al., 2019). The rubric has been employed in prior empirical research (e.g. Lu, Zhu, et al., 2024) and demonstrated acceptable validity. Each criterion was rated on a scale from 1 to 10, yielding a total possible score of 40.

### ***Questionnaire***

A questionnaire was administered to assess participants' writing motivation, feedback literacy, and use of SRL strategies via the online platform Wenjuanxing (<https://www.wjx.cn/>). The questionnaire was presented in Chinese and completed within 10–15 minutes. The writing motivation and SRL strategy use scales were adapted from Bai et al. (2022), and the feedback literacy scale from Yu et al. (2022). Each scale comprised multiple dimensions with several items per dimension. Writing motivation and feedback literacy were measured on a five-point Likert scale (1 = strongly disagree,

5 = strongly agree), whereas SRL strategy use was measured on a five-point scale (1 = never, 5 = always). Detailed information on item numbers, example items, and reliability coefficients is provided in [Appendix B](#).

### **Data collection and analysis**

Data were collected at two time points, at the beginning and end of the ten-week intervention, allowing observation of changes across the instructional period. Additional considerations regarding the research design, including the rationale for not including a control group, are provided in [Appendix C](#).

For RQ1, paired-sample *t*-tests were conducted to examine pre–post changes in writing motivation, feedback literacy, and SRL strategy use. Effect sizes were calculated using Cohen’s *d*. Skewness (–0.313 to 1.365) and kurtosis (–1.471 to 0.542) fell within  $\pm 3$ , indicating acceptable normality.

For RQ2, semi-structured interviews were carried out to explore students’ perceptions and experiences regarding GenAI-supported feedback practice during the three writing tasks. Students were selected based on their initial draft performance in Task 1, assessed using the four scoring criteria described in the ‘Three writing tasks’ section ( $M = 21.70$ ,  $SD = 4.05$ ). Task 1 was chosen because it represented students’ baseline writing ability prior to sustained exposure to GenAI-supported feedback practice, allowing for a more objective grouping that minimised the potential confounding influence of later learning effects.

Students were rank-ordered according to their total Task 1 scores and divided into three approximately equal-sized proficiency groups (tercile split). This resulted in 22 students in the high group ( $M = 26.18$ ), 19 in the medium group ( $M = 21.16$ ), and 20 in the low group ( $M = 17.30$ ). Three students from each group were randomly selected for interviews, yielding nine interviewees (S1–S9). These proficiency groupings were used solely for purposive sampling in the qualitative interviews and were not included in the quantitative analyses.

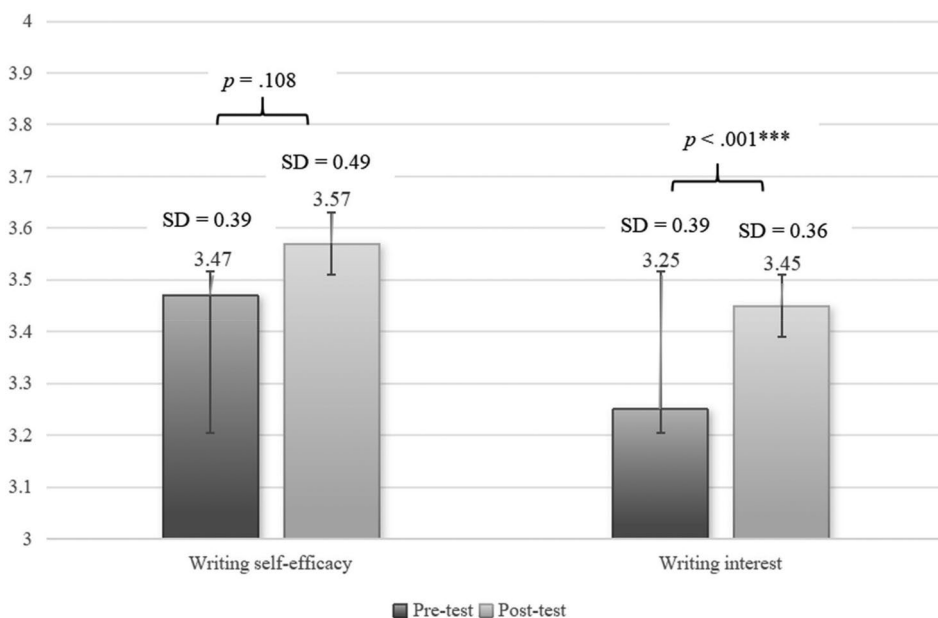
Details of the interview data collection and analysis are provided in [Appendices D and E](#).

## **Results**

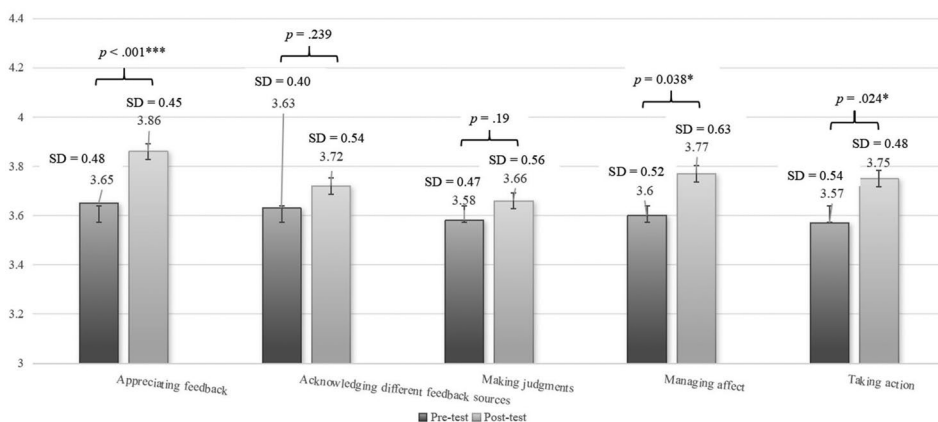
### **Differences in primary variables between pre- and post-test**

To examine RQ1, a paired sample *t*-test was conducted to assess changes in the primary variables from the pre- to post-test. For writing motivation ([Figure 1](#)), writing interest increased significantly from pre-test to post-test,  $M_{\text{pre}} = 3.25$  ( $SD = 0.39$ ),  $M_{\text{post}} = 3.45$  ( $SD = 0.36$ ),  $t(60) = 4.332$ ,  $p < 0.001$ ,  $d = 0.555$ . Writing self-efficacy showed a small, non-significant increase,  $M_{\text{pre}} = 3.47$  ( $SD = 0.39$ ),  $M_{\text{post}} = 3.57$  ( $SD = .49$ ),  $t(60) = 1.630$ ,  $p = 0.108$ ,  $d = 0.209$ .

For feedback literacy ([Figure 2](#)), significant gains were found in appreciating feedback,  $t(60) = 3.781$ ,  $p < .001$ ,  $d = 0.484$ , and managing affect,  $t(60) = 2.125$ ,  $p = 0.038$ ,  $d = 0.272$ , as well as taking action,  $t(60) = 2.321$ ,  $p = 0.024$ ,  $d = 0.297$ . No significant changes were observed for acknowledging sources,  $t(60) = 1.189$ ,  $p = 0.239$ ,  $d = 0.152$ , or making judgments,  $t(60) = 1.325$ ,  $p = 0.190$ ,  $d = 0.170$ .



**Figure 1.** The paired sample *t*-test results for writing motivation. Note. \*\*\* $p < 0.001$ .

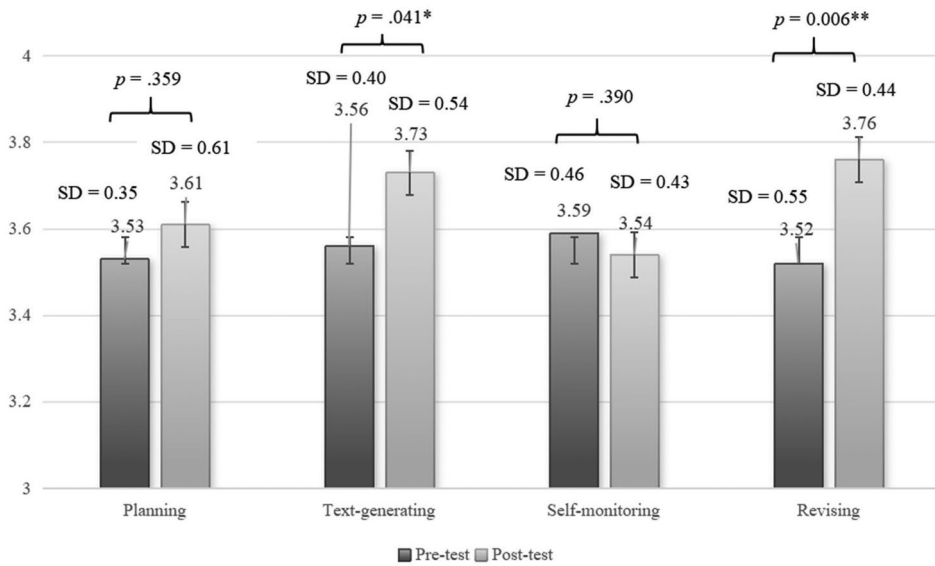


**Figure 2.** The paired sample *t*-test results for feedback literacy. Note. \*\*\* $p < 0.001$ , \* $p < 0.05$ .

For SRL strategy use (Figure 3), significant improvements were found in text-generating strategies,  $t(60) = 2.090$ ,  $p = 0.041$ ,  $d = 0.268$ , and revising strategies,  $t(60) = 2.834$ ,  $p = 0.006$ ,  $d = 0.363$ . Planning strategies showed no significant change,  $t(60) = 0.925$ ,  $p = 0.359$ ,  $d = 0.118$ . A slight, non-significant decrease was observed in self-monitoring,  $t(60) = -0.866$ ,  $p = 0.390$ ,  $d = 0.111$ .

### ***Students' perceptions of GenAI-supported feedback practice***

To explore RQ2, semi-structured interviews were conducted to examine students' perceptions of GenAI-supported feedback practice, focusing on their understanding, preference, and acceptance.



**Figure 3.** The paired sample t-test results for SRL strategy use. *Note.* \*\* $p < 0.01$ , \* $p < 0.05$ .

### Understanding

Regarding understanding, seven out of the nine students reported that they found the feedback provided by Kimi to be easily comprehensible, as it avoided technical jargon and was straightforward. Some students (e.g. S1 and S4 from the high- and medium-proficiency groups) also noted that the feedback was organised in a way that facilitated easy comprehension. However, two students (S6 and S9) from the medium- and low-proficiency groups mentioned that understanding the feedback required a certain cognitive effort. As S9 explained, '*The feedback Kimi gives me is often lengthy. To fully understand it, I need to first spend time comprehending it, and then identify which feedback points are relevant and usable, which takes additional time*'. This suggests that, while the GenAI-generated feedback is understandable, its length and the need to filter relevant information can pose challenges for some students.

### Preference

In terms of preference, students assessed the feedback from both functional and emotional perspectives. Functionally, all nine students expressed positive attitudes regarding Kimi's abilities, highlighting its support for file uploads, rapid feedback, and time-saving features. For instance, S5 (medium proficiency group) commented, '*Once I upload the writing assessment criteria to Kimi, it immediately provides detailed feedback on my writing, saving me time figuring out the criteria. Also, it feels like having a safety net. No matter how I start my writing, Kimi can always help me 'rescue' it*'.

On the emotional side, students from both high- and low-proficiency groups expressed different emotional preferences towards Kimi. Specifically, low-proficiency students felt that Kimi provided emotional support. For example, S8 (low-proficiency

group) shared, *'I find it quite amazing. Although I know I'm interacting with a machine, I feel like it truly understands and listens to what I'm saying, which makes me feel respected. So even though it gives me a lot of negative feedback, I can accept it without feeling uncomfortable'*. In contrast, high-proficiency students valued Kimi's objectivity in providing feedback. For instance, S2 remarked, *'I think Kimi's feedback is well-organised, first pointing out the strengths and then addressing areas for improvement. This structure creates a positive learning experience, and its comments on weaknesses are objective, which makes it feel less like criticism of me as a person'*.

However, not all students expressed strong preference for Kimi. Some, like S6 from the medium-proficiency group, highlighted a more complex feeling. As S6 stated, *'On one hand, I admit that Kimi is really helpful for revising my writing, but on the other hand, after three writing tasks, I'm starting to feel a subtle frustration. I increasingly realise that my writing skills are lacking. I just wish I could write an essay where Kimi doesn't find many obvious mistakes'*.

## Acceptance

Regarding acceptance, students primarily evaluated GenAI-generated feedback from two perspectives: its usefulness and usability. As for the usefulness, students identified three main reasons for considering the feedback beneficial. First, it was seen as helpful in improving the quality of their texts. Specifically, feedback on language and expression was often regarded as valuable, with Kimi's corrections being both precise and thorough. However, feedback related to the content of the text (e.g. organising arguments and adding supporting evidence) was met with mixed reactions. For example, S7 from the low-proficiency group commented, *'Sometimes Kimi suggests adding more evidence, but where am I supposed to get that data from?'*

Second, students found that the feedback was thought-provoking and encouraged reflection. This occurred in two ways: when feedback was general, it prompted students to think about how to make it more actionable and specific; and when students disagreed with the suggested revisions, it encouraged them to reconsider and explore other potential improvements. As S2 (high-proficiency group) stated, *'I don't always accept Kimi's revision suggestions because some of them aren't very practical. However, I think about the underlying revision direction and see if there's room for further improvement in my writing'*.

Third, students appreciated the ability to revisit feedback at any time. For instance, S1 (high-proficiency group) explained, *'When revising for the next writing tasks, I would often look back at the conversation history with Kimi from the previous task to review any issues I might have overlooked. I also try to consciously apply what I've learned from interacting with Kimi, such as argument structures and transitional phrases, in the next task'*.

However, some students raised concerns about the usefulness of the feedback. S5 (medium-proficiency group) noted, *'I've noticed that Kimi's feedback in all three writing tasks has been quite similar, always suggesting that I add more evidence. In the first two tasks, I reconsidered its suggestions, but for the current topic, which I am more familiar with, I feel I've done a good job. Yet, Kimi is still suggesting the same thing, making me question whether its feedback is simply a fixed pattern rather than tailored to my writing'*.

Concerning the usability, students evaluated the ease of converting feedback into revisions. All nine students agreed that the feedback on language, especially spelling and grammar, could be directly applied without much effort. However, when it came to content and logic, students usually needed to further refine the suggestions. Different proficiency levels led to different approaches: high- and medium-proficiency students tended to (1) reflect independently or (2) engage in further discussions with Kimi on specific issues. In contrast, low-proficiency students tended to seek direct answers from Kimi on particular issues.

Nonetheless, some students pointed out that Kimi excelled in guiding the direction of discussions. For instance, S4 (medium-proficiency group) shared, *'After a conversation, Kimi sometimes suggests related topics, which makes you want to click on them. I've done this, and before I knew it, I was no longer discussing my original text. Of course, I acknowledge that small, off-topic conversations can help relax the mind, but I still need to be conscious and purposeful in my communication with Kimi. Otherwise, I might just be led down a different path.'*

## Discussion

Building on the lens of writer independence, the present findings suggest that GenAI-supported feedback practice may reshape independence, understood as learners' autonomy and responsibility in directing their writing development, in a dual way. On the one hand, by providing readily accessible, information-rich, and iterative feedback support, GenAI may strengthen learners' capacities across motivational, feedback literacy, and regulatory domains (e.g. heightened writing interest, enhanced ability to act on feedback, and greater use of text-generating and revising strategies). On the other hand, when evaluative and monitoring functions are partially externalised to GenAI, learners may rely on the tool to identify problems and judge quality, which may reduce opportunities to practice independent evaluation and metacognitive regulation (e.g. limited gains in making judgments and a slight decline in self-monitoring).

Importantly, these patterns imply that writer independence in GenAI-mediated contexts is not simply 'more independence' or 'less independence'. Rather, the autonomy and responsibility central to writer independence become reconfigured: learners may become more willing and more able to act on feedback and regulate their writing, while becoming less autonomously evaluative and self-monitoring if tool reliance replaces internal judgement. The following sections therefore discuss each construct not as separate outcomes, but as complementary process-level indicators that together reveal how writer independence may be strengthened in some dimensions while weakened in others.

### ***The potential influence of GenAI-supported feedback practice on writing motivation***

The paired-sample *t*-test (Figure 1) showed a significant increase in writing interest, while writing self-efficacy improved slightly but not significantly, suggesting that GenAI-supported feedback practice may boost writing enthusiasm but has limited impact on confidence.

Several factors may account for the observed increase in writing interest. Drawing on SDT, one possible explanation lies in the enhancement of students' perceived autonomy, which arises when learners experience a sense of volition and personal choice in their learning (e.g. Graham, 2018; Hidi et al., 2023). In GenAI-supported feedback practices, autonomy is fostered when students independently decide how to engage with GenAI tools, such as whether to accept or disregard GenAI-generated suggestions, rather than following them passively.

At the same time, it is possible that writing was perceived as more manageable when supported by GenAI, which may have reduced initial cognitive strain. If tasks are experienced as less effortful, learners may report greater enjoyment. However, interview data suggest that students' interest was not solely rooted in task simplification, but also in their active role in evaluating and integrating feedback. For instance, S2 (high-proficiency group) described assessing the usefulness and feasibility of GenAI feedback before deciding whether to apply it. This pattern aligns with SDT's claim that autonomy-supportive contexts foster intrinsic motivation, including interest in writing, rather than merely generating enjoyment through reduced task difficulty.

In addition, the increase in writing interest may be associated with enhanced feelings of relatedness as proposed by SDT. Some students (e.g. S8, low-proficiency group) reported feeling 'understood' and 'respected' by the feedback provided by GenAI, which strengthened their emotional connection with the writing task and, in turn, cultivated greater interest in writing.

By contrast, the non-significant change in writing self-efficacy may diverge from previous findings (e.g. Apriani et al., 2024; Huang & Mizumoto, 2024), which reported notable improvements following interaction with GenAI tools. A possible explanation may lie in differences in research design. For instance, in Huang and Mizumoto (2024) study, students collaboratively wrote group essays, meaning that individual exposure to GenAI was relatively limited. In the present study, however, students completed three full individual writing tasks with GenAI in the feedback practices, resulting in more sustained and direct interaction with the tool.

From the perspective of SDT, this result may relate to students' perceived competence, which can fluctuate over time with extended GenAI use. Although a slight improvement was observed, some students (e.g. S6, medium-proficiency group) expressed mild frustration after repeatedly interacting with GenAI, as the feedback occasionally made them more aware of their linguistic limitations. This heightened awareness, while potentially beneficial for reflection, may also introduce a sense of inadequacy. Such feelings can be understood through observation and modelling processes (Bandura, 2021), where learners unconsciously compare their own performance to that of a model. When the perceived gap between the learner and the model becomes too wide, such comparisons may dampen confidence rather than enhance it.

Taken together, these findings suggest that prolonged exposure to GenAI-supported feedback practices may not yield uniformly positive motivational effects. As Rodríguez-Ruiz et al. (2025) cautioned, the motivational consequences of GenAI use are likely multifaceted. From the standpoint of writer independence, the motivational pattern suggests that GenAI may strengthen learners' willingness to take up writing, which is an essential foundation for autonomy in sustained writing development.

However, the modest change in self-efficacy and occasional frustration reported by some students indicate that autonomy-supportive experiences may coexist with moments of self-doubt when learners confront perceived gaps between their current writing and the level of expression reflected in GenAI feedback. This may suggest that fostering independence involves not only sustaining interest but also supporting learners in interpreting GenAI feedback in ways that reinforce realistic and adaptive competence beliefs, rather than triggering maladaptive upward comparisons.

### ***The potential influence of GenAI-supported feedback practice on feedback literacy***

Figure 2 reveals a nuanced pattern of change in students' feedback literacy, indicating both substantial progress and areas of limited development. This finding partly aligns with Rad et al. (2024), who reported generally positive effects of AI integration on feedback literacy. However, the present study reveals a more differentiated pattern: students improved significantly in appreciating feedback, managing affect, and taking action, whereas progress in acknowledging feedback sources and making judgments was comparatively limited.

Appreciating feedback pertains to students' recognition of the value of feedback and their active role in the feedback process. Interview data suggest that direct engagement with GenAI fostered greater appreciation for its immediacy and interactivity. Moreover, as students such as S1 (high-proficiency group) noted, GenAI not only illuminated aspects of their writing that they had previously overlooked but also allowed them to revisit feedback dialogues at any time. This affordance may have helped students reconceptualize feedback as an evolving learning resource rather than a one-off corrective measure, indicating a possible shift from a performance-oriented to a learning-oriented perspective on feedback, a crucial step towards more sophisticated feedback literacy (Molloy et al., 2020).

Managing affect involves students' regulation of emotions when receiving feedback. Beyond feeling 'respected', as reported by S8 (low-proficiency group), several students noted that GenAI's neutral and consistent tone reduced the anxiety and defensiveness often triggered by criticism. From an affective standpoint, such emotionally neutral interactions may help create a psychologically safer feedback space, enabling students to process critical comments with less ego threat and greater openness to improvement (e.g. McGuire, 2025). This suggests that GenAI, by buffering negative emotional responses, may promote more adaptive affective regulation, providing a foundation for sustained engagement in feedback cycles (e.g. Carless & Boud, 2018).

Taking action involves how students respond to feedback. Compared with the 'making judgments', taking action appears to benefit more directly from repeated, concrete opportunities for revision, as each writing cycle requires tangible responses to feedback. In GenAI-supported feedback practices, the quantity and breadth of information provided by the tool (e.g. Lu et al., 2024; Yao et al., 2025) likely expand the range of potential revision options available to students. Even if not all suggestions are directly implemented, this increased diversity of feedback may facilitate greater engagement in active revision, thereby supporting the development of the

taking action dimension. Interview data further indicate that students (e.g. S2, S7, from high- and low-proficiency groups) tend to engage in deliberate decision-making rather than passive acceptance, suggesting that they become more agentic in translating feedback into concrete revision actions.

Acknowledging different feedback sources concerns students' ability to recognise the learning benefits of various sources of feedback. The lack of significant improvement in this area may be attributed to the study design, which exclusively employed GenAI as feedback provider. Without exposure to multiple feedback sources for comparison, students may have limited opportunities to develop their ability to differentiate and assess feedback from diverse perspectives.

Making judgments refers to students' capacity to evaluate their own work and critically assess the quality of feedback received. Unlike the taking action dimension, which showed substantial improvement, gains in judgement-making were more limited, potentially due to two interrelated challenges: cognitive overload from lengthy feedback and students' reliance on GenAI tools.

From the perspective of cognitive load theory, excessive information processing demands can hinder higher-order cognitive tasks such as evaluation and judgement (Sweller, 2011). In GenAI-supported feedback practices, the sheer quantity and length of feedback may have imposed substantial cognitive demands. For instance, S4 (medium-proficiency group) reported that processing lengthy feedback required considerable mental effort, leaving fewer resources available for evaluating its relevance and applicability. While a larger volume of feedback can expand the range of potential revision options (supporting taking action), it does not automatically improve judgement-making; students still need to select, prioritise, and critically appraise the information. In other words, the abundance of feedback may enhance opportunities for action but simultaneously increase the difficulty of making independent evaluative judgments.

Students' reliance on GenAI may further constrain judgement-making. Some, like S5 (medium-proficiency group), indicated that using GenAI reduced the need to internalise evaluation criteria, thereby limiting opportunities to practice autonomous judgement. Moreover, because GenAI often guides feedback in specific directions, students may have been inclined to follow suggestions passively rather than critically engage with them. This combination of cognitive load and tool dependence likely explains why judgement-making improved only modestly, in contrast to taking action, which primarily benefits from the availability of diverse feedback options and tangible opportunities to revise.

Taken together, the differentiated pattern across feedback literacy dimensions highlights a key tension in independence under GenAI-supported feedback practice. The gains in appreciating feedback, managing affect, and taking action suggest that GenAI can strengthen learners' capacity to participate in feedback cycles and translate input into revision, thereby supporting responsibility for improvement. At the same time, limited growth in making judgments indicates that abundant and tool-directed feedback does not automatically cultivate independent evaluative capacity. In GenAI-mediated environments, independence therefore hinges not only on acting on feedback responsibly, but also on sustaining the autonomy to judge feedback quality and relevance. In this sense, being an 'independent writer' in the age of AI entails

retaining evaluative authority over one's writing decisions, rather than merely executing externally generated suggestions. Such evaluative authority may require targeted scaffolding beyond repeated revision opportunities.

### ***The potential influence of GenAI-supported feedback practice on SRL strategy use***

Figure 3 shows that the GenAI-supported feedback practice had mixed effects on SRL strategy use: text-generating and revising strategies improved significantly, planning showed a modest increase, while self-monitoring declined slightly. According to SRL theory, learning involves forethought (goal setting), performance (strategy use and monitoring), and self-reflection (evaluation) (Schunk & Zimmerman, 2011).

The limited improvement in planning may reflect two factors. First, GenAI was not integrated into the pre-writing stage. Second, some students relied on GenAI as a safety net and became less proactive in organising their ideas. As S5 (medium-proficiency) noted, GenAI support reduced their concern about initial writing quality, suggesting lowered involvement in early planning.

Text-generating strategies, part of the performance phase, improved likely because repeated exposure to GenAI feedback across similar IW tasks enabled knowledge transfer (e.g. Nokes, 2009). GenAI served as an additional source of idea recall; for example, S1 (high-proficiency) reported reusing argument patterns and transitions extracted from earlier feedback provided by GenAI. Revising strategies also improved, which is expected since GenAI was directly used during revision activities in this study.

Self-monitoring, which encompasses evaluating one's writing and adjusting processes accordingly, spans both the performance and self-reflection phases. The observed decline in self-monitoring strategies is unexpected, suggesting that prolonged exposure to GenAI may have had a counterproductive effect on students' autonomous regulation. Two potential explanations are proposed. First, the perceived need for self-monitoring may have diminished: as students grew accustomed to GenAI-generated corrections, they may have shifted from a proactive to a reactive revision approach, waiting for GenAI suggestions rather than independently identifying areas for improvement. This reflects a transition from internal self-regulation to external regulation (e.g. Fan et al., 2025).

Second, metacognitive engagement may have been reduced. The convenience of GenAI, particularly its ability to provide immediate and information-rich feedback, may have limited students' opportunities for deliberate error detection and independent evaluation, both of which are critical for developing self-monitoring skills. Importantly, the abundance or complexity of feedback does not necessarily make it easier to act upon; students still need to interpret, prioritise, and decide how to apply it. Moreover, when students receive lengthy or information-dense feedback, cognitive resources may be preferentially allocated to implementing revisions rather than critically monitoring their performance or evaluating feedback quality. This redistribution of attention may highlight a potential trade-off between execution-oriented engagement (taking action) and reflective, evaluative regulation in GenAI-mediated contexts. Notably, this pattern corresponds with the limited improvement observed in the making judgments dimension of feedback literacy, suggesting that the cognitive and

metacognitive demands of evaluating feedback were not fully met when students relied heavily on GenAI.

Taken together, the SRL strategy pattern suggests that GenAI-supported feedback practice may differentially shape regulatory processes. While execution-oriented regulation, such as text-generating and revising strategies, was strengthened through repeated interaction with GenAI, reflective regulation, particularly self-monitoring, showed slight decline when monitoring functions were increasingly externalised to the tool. This pattern does not necessarily indicate a loss of independence per se, but rather a redistribution of regulatory effort: learners may invest more cognitive resources in implementing revisions while allocating less to internally detecting problems or evaluating their own performance.

From the perspective of writer independence, this redistribution invites a reconceptualization of what independence entails in GenAI-mediated writing contexts. When GenAI assumes part of the monitoring and judging functions traditionally associated with self-regulation, independence can no longer be equated with performing all regulatory processes unaided. Rather, independence may lie in learners' capacity to retain autonomous regulatory authority over their writing processes, including deliberate oversight of when and how external inputs are integrated into ongoing planning, drafting, and revision. In this sense, independent writers are not those who avoid technological assistance, but those who sustain metacognitive control over tool-mediated regulation rather than allowing regulatory processes to be fully externalised.

## Conclusions

This study advances understanding of how GenAI-supported feedback practice shapes the development of independent L2 writers in academic writing contexts by clarifying a central tension. While GenAI may help learners feel and act more independently by strengthening their willingness to engage in writing and supporting feedback uptake and revision-related regulation, it may simultaneously reduce opportunities to think and evaluate independently if monitoring and judging functions are increasingly externalised to the tool. In this sense, independence in GenAI-mediated L2 writing contexts, understood as learners' autonomy and responsibility in directing their writing development, is not defined by the absence of technological support, but by their capacity to retain ultimate evaluative and regulatory authority while engaging with such support. Independence is therefore reconfigured rather than diminished in the age of AI: it resides not in rejecting assistance, but in governing its use in ways that sustain autonomy and responsibility.

The findings provide several implications for integrating GenAI into L2 writing instruction. To support writing self-efficacy, instructors should clarify both the strengths and limitations of GenAI tools and guide students in using them appropriately rather than treating GenAI outputs as normative standards. For example, teachers might explicitly model how GenAI suggestions are generated (e.g. by prompting the tool with varying levels of detail) and discuss why certain outputs appear fluent yet may lack contextual appropriateness or critical depth. Such guided analysis can help students understand that GenAI outputs are generated responses shaped by prompts

and training data, rather than ideal benchmarks against which their own writing should be judged.

For feedback literacy, particularly in recognising multiple feedback sources, teachers may combine GenAI-generated feedback with teacher and peer feedback to provide diverse evaluative perspectives and reinforce their complementary roles (e.g. Lu et al., 2024). For example, students may categorise feedback from different sources (GenAI, peers, teacher) and reflect on the distinctive strengths of each source to support more strategic feedback use.

To strengthen judgement-making, instructors may design tasks that require students to compare GenAI outputs with evaluation criteria and their own texts. For instance, students could evaluate a GenAI-generated paragraph using a rubric before revising their drafts and then reflect on similarities and discrepancies. Such activities may cultivate critical evaluation skills and reduce unreflective acceptance of GenAI suggestions.

Regarding SRL strategy use, explicit instruction in goal-setting and text organisation can strengthen planning during the forethought phase. For example, teachers may guide students to set specific revision goals before consulting GenAI (e.g. focusing on argument clarity or paragraph coherence) rather than requesting broad feedback.

To enhance self-monitoring, both awareness and structured practice are needed. Teachers might incorporate short self-assessment checklists before submitting drafts to GenAI (e.g. Yan, 2020), embed reflective prompts during GenAI interactions (e.g. 'Why is this suggestion helpful?'), and require students to explain why they accept or reject GenAI suggestions (e.g. Teng, 2025). Such practices help sustain internal regulation while benefiting from external support.

Finally, differentiated support may be necessary, as lower-proficiency students may experience cognitive overload when confronted with extensive GenAI feedback. Teachers can scaffold feedback uptake by helping students prioritise suggestions (e.g. addressing global issues before local language corrections) or modelling how to identify the most actionable feedback points.

Some limitations should be noted. First, the study involved education-major students from a single Chinese university, and the sample was predominantly female. These characteristics may limit the generalisability of the findings to other disciplinary, institutional, or demographic contexts. Future research should include more diverse disciplines, institutions, and more balanced demographic compositions to examine the robustness of the observed patterns.

Second, as all participants received the same instructional program, no control group was established. While this design aligns with the study's developmental focus on examining changes in students' writing motivation, feedback literacy, and SRL strategy use under GenAI-supported feedback practice, it limits the ability to attribute these changes solely to GenAI use. Future research could incorporate control groups or comparative designs to better isolate the contribution of GenAI from other instructional factors.

Third, although the two-wave design allowed observation of pre-post changes, it may still overlook nonlinear developmental patterns (e.g. fluctuations in self-monitoring). In addition, although writing drafts and GenAI chat logs were used for qualitative triangulation, the study did not conduct systematic revision-tracking analyses linking

specific feedback to corresponding textual changes, nor include finer-grained process indicators (e.g. detailed revision histories or time-stamped interaction data). Higher-frequency longitudinal designs and more fine-grained process-tracing approaches (e.g. revision mapping, uptake coding, or keystroke tracking) would better capture long-term developmental trajectories and the dynamics of feedback uptake in GenAI-supported writing contexts.

Fourth, while the present study focused on learners' capacities to process and enact feedback once provided, it did not examine their ability to actively elicit feedback in GenAI-mediated environments. In such contexts, students' ability to formulate, refine, and strategically adjust prompts may shape both the feedback generated and how it is subsequently interpreted and used. Future research could therefore extend current conceptualizations of feedback literacy by incorporating prompting-related capacities alongside feedback processing dimensions to better capture the evolving demands of technology-enhanced writing practices.

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No potential conflict of interest was reported by the author(s).

## Conflict of Interest

The authors declare that no generative AI or AI-assisted technologies were used in the preparation of this manuscript, except for minor language polishing and grammatical checking.

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## Data availability statement

The data that support the findings of this study are available from the first author, upon reasonable request.

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## Appendix A

### *Preparatory instruction and procedures for GenAI-supported feedback practice*

The SRL strategy instruction consisted of three parts. The first part involved explaining SRL strategies and their connection to writing evaluation criteria, emphasising the students' awareness of the importance of these strategies in improving their writing. The second part involved group discussions, where participants, in groups of four to five, shared their understanding of SRL strategies and discussed any previous experiences with using them. In the third part, groups shared their discussions with the larger class.

Regarding the Kimi training, the goal was to help participants understand how to craft effective prompts to obtain valuable feedback. Kimi is a large language model that provides draft-level feedback through a Chinese-language interface, supporting multimodal inputs such as text, images, and documents. Students can submit full drafts and receive context-sensitive suggestions to guide revision. Compared with other widely available GenAI tools (e.g., ChatGPT), Kimi was selected for its stable access within mainland China, compatibility with local user preferences, and seamless integration into document-based writing tasks.

The training was structured in three phases. In the first phase, participants learned prompt creation strategies aligned with guidelines proposed by Atlas (2023), including providing contextual information, assigning specific roles, detailing tasks, and specifying expected outcomes. Concrete examples were provided. For instance, an initial prompt (originally in Chinese) could be translated as:

*"You are an experienced English writing instructor. Please refer to the evaluation rubric I have uploaded and provide feedback on my writing, indicating areas you think could be improved."*

A follow-up prompt could be translated as:

*"Based on the feedback you provided ('XXXX'), please further clarify how it should be interpreted."*

These examples illustrated how prompts could be structured to elicit detailed and actionable feedback from Kimi. In addition, students were encouraged to use prompts flexibly, particularly by crafting follow-up prompts to refine and expand upon the initial feedback.

The second phase involved hands-on practice with Kimi. In groups of four to five, participants were given a writing sample along with its corresponding evaluation criteria and applied the prompt strategies from the first phase to generate feedback. Special emphasis was placed

on using the evaluation criteria effectively, including uploading the criteria to Kimi before starting the evaluation to ensure clarity and precision in the feedback received.

The third phase involved sharing experiences and challenges encountered during the Kimi evaluation process. Importantly, although guidance was provided during the training sessions, students retained full autonomy in their actual use of Kimi during the feedback practice. They could freely decide how to phrase prompts, which suggestions to adopt, and how to incorporate feedback into their writing. This approach allowed the study to capture authentic interaction patterns and examine how writing motivation, feedback literacy, and SRL strategy use may change in a semi-structured yet open learning environment.

Throughout the ten-week program, participants completed three writing tasks (in Weeks 3, 6, and 9). Initial drafts were composed independently in class without the use of GenAI. Students were instructed that GenAI was to be used only during the designated feedback-and-revision stage. After submitting their drafts, participants spent 40 minutes interacting with Kimi, in either Chinese or English, to obtain feedback and revise their writing. For these three focal writing tasks, GenAI served as the primary source of feedback, and no additional formal teacher or peer feedback was provided during the revision stage. Both the in-class drafting and the subsequent GenAI-supported feedback-and-revision sessions were conducted in supervised classroom settings. During the GenAI-supported feedback-and-revision sessions, the instructor circulated to provide technical assistance related to accessing and operating Kimi, as well as procedural clarification. This in-class supervision helped ensure adherence to the designated use of GenAI during the intervention.

## Appendix B

The scales used in the present study.

| Scale              | Dimension                                | Number of items | Example item                                                                                                              | Pre-test $\alpha$ / Posttest $\alpha$ |
|--------------------|------------------------------------------|-----------------|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| Writing Motivation | Self-efficacy                            | 5               | "When I put in effort, I believe I can produce a good piece of writing"                                                   | 0.736/0.874                           |
|                    | Writing interest                         | 4               | "Writing in English is an enjoyable activity"                                                                             | 0.720/0.925                           |
| Feedback Literacy  | Appreciating feedback                    | 10              | "I think that feedback can enhance my English writing skills"                                                             | 0.902/0.915                           |
|                    | Acknowledging different feedback sources | 5               | "I recognise that feedback from various sources—such as teachers, peers, and AI—provides distinct learning opportunities" | 0.751/0.910                           |
|                    | Making judgments                         | 5               | "I think that standards and criteria are essential for evaluating my English compositions"                                | 0.857/0.952                           |
|                    | Managing affect                          | 3               | "I can stay emotionally balanced and avoid defensiveness when receiving critical feedback"                                | 0.899/0.920                           |
|                    | Taking action                            | 5               | "I can apply feedback to improve my English writing"                                                                      | 0.906/0.922                           |
| SRL Strategy Use   | Planning strategies                      | 4               | "I consider the main ideas I want to express"                                                                             | 0.784/0.909                           |
|                    | Text-generating strategies               | 4               | "I recall ideas I have learned from past writing experiences and apply them to my own compositions"                       | 0.766/0.889                           |
|                    | Self-monitoring strategies               | 5               | "I repeatedly review the evaluation criteria to help generate new ideas"                                                  | 0.778/0.824                           |
|                    | Revising strategies                      | 4               | "I reorganise my ideas"                                                                                                   | 0.803/0.888                           |

## Appendix C

### *Considerations regarding the research design*

It is important to note that the ten-week academic English writing program was designed to provide all participating students with access to GenAI-supported feedback practices within a naturalistic instructional setting. As a result, no formal control group was established. This design aligns with the developmental focus of the present study, which aims to explore whether and how students' writing motivation, feedback literacy, and self-regulated learning (SRL) strategy use change during participation in GenAI-supported feedback practices, rather than to rigorously establish causal effects of GenAI on these variables.

To enhance the interpretability of the observed changes, several measures were implemented. First, both quantitative and qualitative data were collected from multiple sources, including pre- and post-questionnaires administered in Weeks 1 and 10, students' writing products, GenAI chat logs, and semi-structured interviews with a subset of nine students. Second, all participants engaged in the same instructional program with consistent writing tasks and structured opportunities to interact with GenAI during feedback activities, thereby minimising variation in instructional exposure. Third, the alignment between writing tasks and feedback sessions allowed for a closer connection between the feedback received and subsequent changes in students' writing-related variables.

Together, these design features provide a more informed basis for interpreting whether and how GenAI-supported feedback practices relate to observed changes, while also offering rich insights into students' experiences and perceptions of these practices.

## Appendix D

### *Interview data collection and analysis*

The interviews, conducted in Chinese, followed a structured protocol that prompted participants to (1) reflect on their experiences with the three writing tasks and the strategies they employed, (2) discuss their interactions with Kimi during the tasks, (3) share their perceptions of Kimi's role in supporting their writing (including understanding, preferences, and acceptance), and (4) describe their revision processes. The interview questions (e.g., "How well did you understand the feedback provided by Kimi?") were adapted from existing research (e.g., Lu et al., 2024; Yao et al., 2025) to encourage detailed responses. Each interview lasted approximately 40 minutes, was audio-recorded with participant consent, and transcribed verbatim for analysis by the first author and a research assistant with a master's degree in education.

The data were analysed using NVivo 12.0. Two independent coders conducted the analysis, resolving any discrepancies through discussion. Coding primarily focused on students' perceptions of GenAI-supported feedback practice, specifically their understanding, preferences, and acceptance, in line with Lu et al. (2024). For example, statements such as "*I liked the GenAI-supported feedback activities; they felt novel*" were coded under preference. Coders also remained attentive to any unexpected insights emerging from the interviews. In addition to interview data, students' writing drafts and Kimi chat logs were incorporated for data triangulation.

Triangulation was conducted by cross-referencing interview accounts with the corresponding Kimi chat interactions and students' pre- and post-revision drafts. When students described how they used Kimi during feedback uptake and revision, coders examined relevant segments of the chat logs to identify the type of suggestions provided and compared the paired drafts to determine whether and how those suggestions were taken up.

For instance, in the subsection "Students' perceptions of GenAI-supported feedback practice," S7 (low-proficiency group) remarked, "*Sometimes Kimi suggests adding more evidence, but where am I supposed to get that data from?*" This comment suggests that the student questioned the feasibility of certain Kimi-generated suggestions. Consistent with this account, the corresponding chat log showed that Kimi recommended "introducing authoritative data or research findings."

However, this suggestion was not incorporated in the student’s revised draft, as evidenced in the draft comparison (see [Appendix E](#) for the interview excerpt, the linked chat segment, and the associated revision). This cross-referencing enabled us to examine the alignment between students’ reported perceptions, their interaction with GenAI, and their actual revision decisions, thereby grounding our interpretations in observable process and product data.

Briefly, triangulating these data with interview responses enhanced the credibility of our findings in two ways. First, it strengthened the methodological rigour of the qualitative analysis by linking self-reports with process-level traces (chat logs) and textual outcomes (draft revisions). Second, it provided contextual insights that helped interpret potential reasons underlying the observed changes in the focal variables.

Appendix E

Example of triangulation

| Interview excerpts                                                                                  | Kimi chat log (original)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Kimi chat log (Translated into English)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Students’ writing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Sometimes Kimi suggests adding more evidence, but where am I supposed to get that data from?</i> | <div>2. Citation and Synthesis (引用与综合)</div> <div>得分: 3/10</div> <div>问题:</div> <ul style="list-style-type: none"><li>全文没有引用任何外部资料、研究、统计数据或他人观点。</li><li>没有展示对不同观点的综合分析。</li></ul> <div>改进建议:</div> <ul style="list-style-type: none"><li>引入一些权威数据或研究结果。例如:<ul style="list-style-type: none"><li>哪些专业的就业率较高?</li><li>兴趣与职业满意度之间是否有研究支持?</li></ul></li><li>引用不同人的观点 (例如学生、家长、雇主) 来展示问题的复杂性。</li></ul> <div>示例补充:</div> <div>According to a 2022 survey by the National Association of Colleges and Employers, graduates in computer science and engineering have the highest employment rates within six months of graduation. However, a study by Harvard Business Review also found that employees who are passionate about their work are 30% more likely to stay in their jobs long-term.</div> | <div><b>2. Citation and Synthesis</b></div> <div><b>Score: 3/10</b></div> <div><b>Problems:</b></div> <div>The essay does not cite any external sources, research studies, statistical data, or others’ viewpoints.</div> <div>It does not demonstrate synthesis of different perspectives.</div> <div><b>Suggestions for Improvement:</b></div> <div>Introduce some <b>authoritative data or research findings</b>, such as:</div> <div>Which majors have higher employment rates?</div> <div>Is there research supporting the relationship between interest and job satisfaction?</div> <div>Cite different perspectives (e.g., students, parents, employers) to show the complexity of the issue.</div> <div><b>Example Addition:</b></div> <div>According to a 2022 survey by the National Association of Colleges and Employers, graduates in computer science and engineering have the highest employment rates within six months of graduation. However, a study by Harvard Business Review also found that employees who are passionate about their work are 30% more likely to stay in their jobs long-term.</div> | <div><b>Initial draft:</b></div> <div>...</div> <div>University is not only for a diploma; it is also a place to grow mind. When interest and employability meet, our study and future career will both be happier....</div> <div><b>Revised draft:</b></div> <div>...</div> <div><u>University should not be seen only as preparation for employment, but also as an important stage for self-development. By balancing interest with realistic career prospects, we can make choices that help us grow and prepare for the future.</u></div> |